

B5 MAX

Metal Bath Heating & Cooling Instrument

Software Operation Manual

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1 Overview

1.1 Purpose of This Manual

The purpose of this manual is to enable users to correctly, safely, and effectively operate the Metal Bath Heating & Cooling Control System, achieving precise temperature control to meet laboratory and industrial application requirements.

1.2 Icon Definitions

The following table lists the icons used in the interface and their respective states:

Icon	Meaning	States
Manual Setting	Manual temp setting mode	Highlighted / Normal / Disabled
Program Control	Program control execution	Highlighted / Normal / Disabled
Program Edit	Program editing interface	Highlighted / Normal / Disabled
Run Curve	Real-time temperature chart	Highlighted / Normal / Disabled
System Settings	System configuration menu	Highlighted / Normal / Disabled
Run	Start operation	Highlighted / Normal
Pause	Pause operation	Highlighted / Normal / Disabled
Stop	Stop operation	Highlighted / Normal / Disabled
Common 4/37/65/85C	Quick set temperature	Normal / Highlighted
Rate Mode	Basic / High / Extreme	Normal / Highlighted
Confirm / View / Delete	Action buttons	Normal / Highlighted
Previous / Next / Return	Navigation	Normal
Low Temp Preservation	Post-run cool storage	Normal / Highlighted
Power-On Preset	Auto preset on boot	Normal / Highlighted
Hot Lid / Alarm	Lid and alarm mgmt	Normal / Highlighted
Data Export / Self-Test	Export and diagnostics	Normal / Highlighted
Log / Language / Cal	Logging and calibration	Normal / Highlighted
Settings	System settings	Normal / Highlighted

Note: "Highlighted" means active; "Normal" means available; "Disabled" means unavailable. The info icon displays messages to read before proceeding.

2 System Login

After powering on, the screen first displays the boot screen (~3 seconds), then automatically enters the power-on self-test sequence. The system checks whether temperature sensors are working properly within 10 seconds. Once the self-test passes, it enters the Manual Setting page.

[Figure 2-1] Power-On Self-Test Page

After successful self-test, as shown below, the system enters the Manual Setting main page and is ready for operation.

[Figure 2-2] Manual Setting Main Page

3 Manual Mode

3.1 Manual Mode Page Overview

The Manual Mode page is the main operating interface. Users can set temperature, time, rate mode, and control device start/stop from this page.

[Figure 3-1] Manual Mode Page

- **Module Model:** Displays the identified module model (auto-detected via hardware ID pins, supports up to 16 module types).
- **Rate Mode:** Select from Basic, High, or Extreme three-level heating/cooling rates.
- **Common Temperature:** Quick buttons for 4C, 37C, 65C, 85C.
- **Return to Room Temp:** One-button recovery to 23C. Stops at 23+-1C, button pops, buzzer beeps.
- **Set Temperature:** Target temp range 4-100C, precision 0.1C.
- **Set Time:** Holding time (HH:MM:SS), max 99:59:59. Zero = continuous run.
- **Real-Time Temp:** Current sample zone temperature.
- **Hold Countdown:** Auto-countdown after reaching set temp.
- **Run Time:** Cumulative device run duration.
- **Run/Pause/Stop:** Three buttons for start/stop control.

Note: Both Set Temperature and Set Time have power-off memory. After restart, the device restores the last set values.

3.2 Temperature Setting

Click the "Set Temperature" input to open the numeric keypad. Enter the desired target temperature and press "Confirm." Range: 0-105C; out-of-range values trigger a warning.

[Figure 3-2] Temperature Input Keypad

3.3 Time Setting

Click "Set Time" to open the time setting interface. Set hours, minutes, seconds individually. Max: 99:59:59. When set to 0, the device runs continuously (no countdown).

3.4 Rate Mode

Click the "Rate Mode" dropdown to select one of three heating/cooling rates:

Mode	Heating Rate	Cooling Rate	Application
Basic	~2C/min	~0.5C/min	Low rate-change requirement
High	~5C/min	~1C/min	Rapid temperature change needed
Extreme	Unlimited	Unlimited	Fastest speed to target

3.5 Common Temperature Shortcuts

The Manual Mode page provides 4 common temperature shortcut buttons: 4C, 37C, 65C, 85C. Clicking sets the target automatically. When the set temperature is not one of these, buttons remain unselected.

3.6 Return to Room Temperature

Click "Return to Room Temperature" to auto-control to 23C. When temperature reaches 23+-1C, it auto-stops, the button pops, and the buzzer beeps once. Suitable for safe sample removal after experiments.

During "Return to Room Temp" operation, clicking "Stop" cancels the function. The fan continues for 10 seconds before stopping.

4 Program Control

4.1 Program Control Page Overview

The Program Control page executes preset multi-step temperature control programs for automated operation.

[Figure 4-1] Program Control Page

- Set Temperature: Target temp of current step (set in Program Edit).
- Set Time: Holding time of current step (HH:MM:SS).
- Rate Mode: Heating/cooling rate of current step.
- Program Name: Dropdown in top-right to select program.
- Current Step: Step number being executed.
- Real-Time Temp: Current sample zone temperature.
- Hold Countdown: Countdown after reaching step temp.
- Run Time: Cumulative program control run time.
- Run/Pause/Stop: Program start/stop control.

Note: In program control mode, when step countdown ends, it auto-switches to the next step. After all steps complete, the device auto-stops and the buzzer alerts.

4.2 Program Editing

4.2.1 Program Library Page

[Figure 4-2] Program Edit Page (Program Library)

On the Program Edit page, select the program to run and view detailed steps. Max 5 programs, each with up to 10 steps. Programs have power-off memory.

- Click "View" to see program details.
- Click "Delete" to remove the program.

4.2.2 Program View

Click "View" to enter the Program View page, showing all step details (target temp, holding time, rate mode). Click "Program Edit" to modify. The program name can be changed via dropdown.

[Figure 4-3] Program Edit - View Page

4.2.3 Program Edit Page

[Figure 4-4] Program Edit - Edit Page

(1) Three add (+) buttons on the left. When pressed, the corresponding row becomes editable. When released, content is deleted and steps move forward. (Steps add sequentially.)

(2) Steps paginated with "Previous"/"Next". Max 10 steps per program.

(3) Click "Save" after editing. If time or temperature = 0, save is blocked with a popup. Temperature must be 4-100C.

Note: Click "Cancel" or "Return" to go back to the Program Library page.

5 Run Curve

[Figure 5-1] Run Curve Page

The Run Curve page displays real-time temperature trends for intuitive understanding of the temperature control process.

- Vertical axis: Temperature (C); Horizontal axis: Time (min).
- Reference lines for common temperatures: 4C, 37C, 65C, 85C, 100C.
- Records up to 12,800 data points; each page shows ~1 hour (720 points).
- Use Page X / Previous / Next to browse historical trends with quick jump.
- Real-time curve update during operation; review after stopping.

Note: Page navigation is enabled only after the current page finishes drawing.

6 System Settings

6.1 System Settings Page Overview

[Figure 6-1] System Settings Page

(1) When entering System Settings during operation, all function spheres except "Settings" and "Calibration" are disabled.

(2) After entering System Settings > Settings and Calibration menus, Manual Mode and Program Edit menus are dimmed.

The System Settings page uses a circular icon layout with these sub-modules:

No.	Module	Description
1	Low Temp Preservation	Auto-enter low temp mode after run
2	Power-On Preset	Auto pre-heat/cool to set temp on boot
3	Self-Test	Re-run boot self-test sequence
4	Language Switch	CN/EN interface (not yet active)
5	Hot Lid Mode	Independent hot lid temp control
6	Alarm Information	View historical alarm records
7	Calibration	Temperature compensation calibration
8	Settings	RTC time setting, factory reset
9	Data Export	USB/SD card data export
10	Log	View run log records

Note: A password is required to enter the Calibration page from System Settings.

6.2 Low Temperature Preservation

[Figure 6-2] Low Temperature Preservation Settings

After the main experimental program finishes, the instrument automatically lowers the module to a specific temperature (typically 4C) and maintains it to prevent sample degradation.

- Low Temp Switch: ON/OFF. When ON, enters low temp mode after run.
- Preservation Temp: Setting range 4-20C.
- Preservation Time: Duration (HH:MM:SS). Auto-exit when time expires.

Operation Logic:

- Auto-Transition: Seamless switch to preservation after main program ends (Manual or Program Control).
- Timed Mode: If time is set (e.g., 60 min), instrument maintains low temp then auto-stops.
- Continuous Mode: If no time (0/infinite), instrument maintains low temp until manually stopped.

6.3 Power-On Preset

[Figure 6-3] Power-On Preset Settings

Auto-executes a preset constant-temperature program on startup, quickly reaching standby temperature (e.g., 25C or 37C), reducing wait time.

- Preset Switch: ON/OFF.
- Preset Temperature: 4-100C. When ON, starts temperature control on power-up.

Operation Logic:

- Auto-Start: After self-test, if Preset is ON, runs directly without user intervention.
- Priority: Lower than user main program. Starting a main program terminates Preset immediately.

6.4 Hot Lid Mode

[Figure 6-4] Hot Lid Mode Settings

Independently controls hot lid temperature to prevent condensation on sample lid inner surfaces (e.g., PCR).

- Hot Lid Switch: ON/OFF.
- Temperature Difference: Hot lid above module temp by 5-20C, max 110C.
- Note: Hot lid disabled when module temp is below 20C.

Operation Logic:

- Independent Control: On/off independent of module control. Set lid target independently.
- Anti-Condensation: Keeps tube top hotter than reagent, preventing cap condensation and ensuring concentration stability.
- Associated Behavior: Pre-heat lid before main program. May auto-close during low temp mode (<15C).

Note: In Hot Lid Mode: Target Temp = Current Target - Lid Diff x 5%.

6.5 Log

[Figure 6-5] Log Page

(1) Each page: 5 rows with serial no., start time (HH:MM:SS), end time (HH:MM:SS), run mode (Manual/Program), and details.

(2) Click "Details" icon for detailed run info of that entry.

(3) Max 100 log entries, paginated with Previous/Next.

(4) Logs have power-off memory.

(5) Click "Clear" to clear all entries. (May take some time.)

Note: Log Detail has Manual and Program Control sub-pages.

6.6 Alarm Information

[Figure 6-6] Alarm Information Page

(1) Max 100 alarm entries, paginated with Previous/Next.

(2) Alarm info has power-off memory.

(3) Click "Clear" to clear all entries. (May take some time.)

Alarm Type	Description
Temp Sensor Fault	No valid temp data during self-test
Temperature Anomaly	No temp change for long period, target not reached
Module Not Installed	Module ID not recognized
Communication Fault	Serial touch screen communication error
Memory Fault	FRAM read/write error
Hot Lid Not Detected	Hot lid module not connected

6.7 Self-Test

[Figure 6-7] Self-Test Page

Click "Self-Test" to re-run the self-test sequence. Checks if temperature sensors read valid data within 10 seconds. Returns to System Settings on success. Fan speed at 50% during self-test.

6.8 Calibration (Temperature Compensation)

Compensates temperature measurements for better accuracy. Supports 4 compensation value sets for 4 temperature ranges:

Range	Temperature Range
Low Temp	0-15C (incl. 0C)
Normal Temp	15-55C (incl. 15C and 55C)
Medium Temp	55-85C
High Temp 1	85-100C (incl. 85C)
High Temp 2	100-105C (incl. 100C)

Users adjust compensation based on a standard thermometer for accuracy.

- 5mlx8, 15mlx8, 50mlx4 are large modules - use large module compensation.
- 0.6mlx24, 1.5mlx24, 2.0mlx24 well plates are small modules - use small module compensation.

6.9 Settings

- RTC Time Setting: Set real-time clock (Year/Month/Day/Hour/Minute) for accurate timestamps.
- Factory Reset: Restore all parameters to defaults.

6.10 Data Export

- USB Export: Export run log data to USB (Excel format) for analysis.
- SD Card Auto-Save: Periodic auto-save every 5 minutes.

6.11 Product Information

[Figure 6-8] Product Information Page

Displays basic product information:

Item	Content
Product Name	B5 MAX Intelligent Metal Bath
Product Model	B5 MAX
Software Version	B5.MAX-V3.2.2
Screen Version	V6.6.2
Time Setting	Set machine date and time

7 Operation Instructions

7.1 Run Mode Introduction

The Metal Bath Heating & Cooling Instrument supports two main operating modes:

- Manual Mode: On the Manual Setting page, set target temp and holding time, click "Run." Click "Stop" or auto-stop when time expires.
- Program Control Mode: Select/edit a program, set parameters for each step, return to Program Control page, click "Run." Auto-stop when all steps complete.

7.2 Manual Mode

7.2.1 Preparation

- Confirm mode: Manual Setting page
- Set target temperature (4-100C)
- Set holding time (0 = continuous)
- Select rate mode (Basic/High/Extreme)

7.2.2 Operation Method

Click "Run" - the device starts heating/cooling to target temp. The Run button lights up; Stop turns off.

When temperature reaches the set value, hold countdown starts automatically.

When countdown ends, buzzer beeps 3 times and device stops.

During operation, click "Pause" to pause; click "Run" to resume.

Click "Stop" at any time to stop.

Important Notes:

- If a temperature anomaly occurs, device auto-stops and records alarm info.
- Power-Off Resume: After unexpected power failure, device auto-resumes from breakpoint on restart.
- If module is not installed, display shows "Module Not Detected" and Run/Pause/Stop are grayed.
- During operation, Set Temperature and Set Time inputs are locked; Rate Mode cannot be changed.

7.3 Program Control Mode

7.3.1 Preparation

- Confirm mode: Program Control page
- Select program in Program Edit page
- Confirm all step parameters are correctly set

7.3.2 Operation Method

Click "Run" - executes Step 1, heating/cooling to its target temperature.

When step temp is reached, hold countdown starts. On completion, auto-switch to next step with one buzzer beep.

Current step number, target temp, and countdown update in real time.

After all steps, buzzer beeps 3 times and device stops.

Important Notes:

- Click "Stop" anytime to halt.
- After stopping and restarting, step/temp/time revert to Step 1 values.
- Temperature anomalies trigger auto-stop and alarm recording.
- Power-Off Resume: Auto-resumes from breakpoint (including step and remaining time).
- Program name is locked during program control operation.
- Lower heating plate must not exceed 108C (physical limit).

8 Technical Specifications

Category	Parameter
Main Controller	STM32F407VETx (168MHz Cortex-M4F)
Operating System	FreeRTOS 10.x Real-Time OS
Display	7-inch DGUS Serial Touch Screen
Temperature Sensor	PT1000 Platinum RTD (4 channels)
ADC Precision	AD7124 24-bit High-Precision ADC
Memory	FRAM FM24V05 64KB (non-volatile)
Control Algorithm	Dual Closed-Loop Cascade PID (Incremental + Positional)
Communication	Modbus, DGUS Serial Screen Protocol
Temperature Range	0C ~ 105C

Category	Parameter
Heating	3 modes: Full Power (>5C/min), High (5C/min), Med (2C/min). Heating time: <=15min (25C to 100C)
Cooling	3 modes: Full Power (>=2C/min), High (>=1C/min), Med (>=0.5C/min). Cooling time: <=15min (100C to 25C)
Modules	Multiple tube specs (0.6, 1.5, 2, 5, 15, 50 mL). Multi-well plates. Tool-free swap.
Lids	Standard lid (precision control). Optional hot lid (1.5/2mL anti-condensation).
Programs	Up to 50 programs, 10 steps each
Log Capacity	100 run logs + 100 alarm records
Temp Curve	Up to 12,800 data points
Code Scale	~21,000 lines of C code

Important Notes

1. Before turning off the instrument, stop device operation first, wait for the temperature to return to room temperature, then turn off the power switch.
2. Never restart the instrument immediately after turning it off. Improper operation may affect the instrument service life.

QR Code for Instructions

Scan the QR code below using WeChat or a browser to view the instrument operation instructions:

[QR Code Area - Scan with WeChat or Browser]